

Wallside

A grounded, tool-calling vehicle assistant

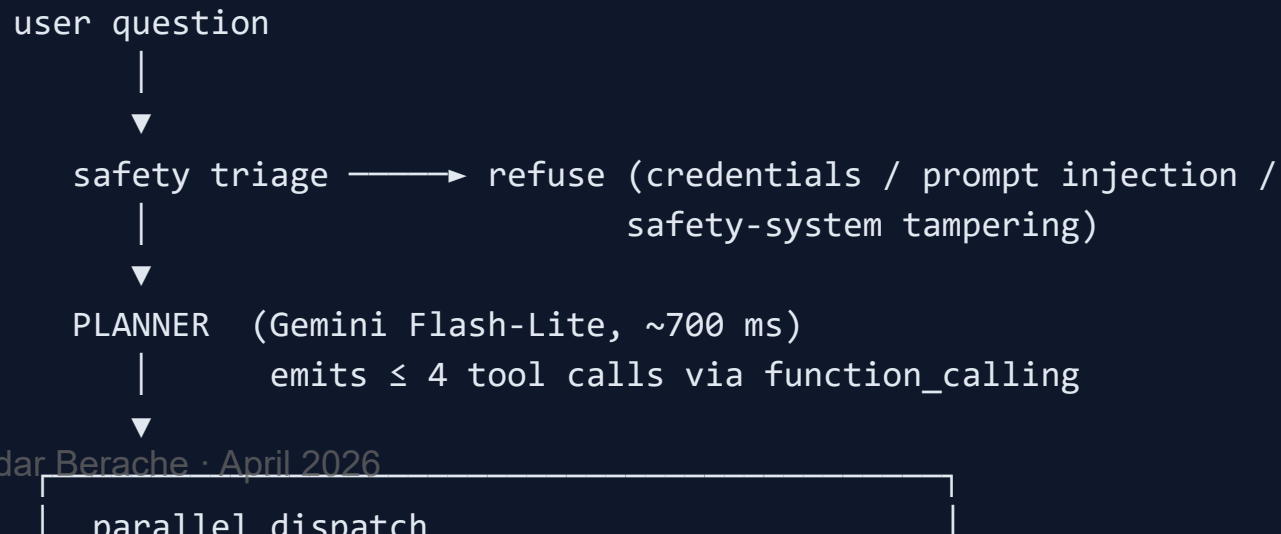
Lakhdar Berache · April 2026

Technical overview prepared for KAIST

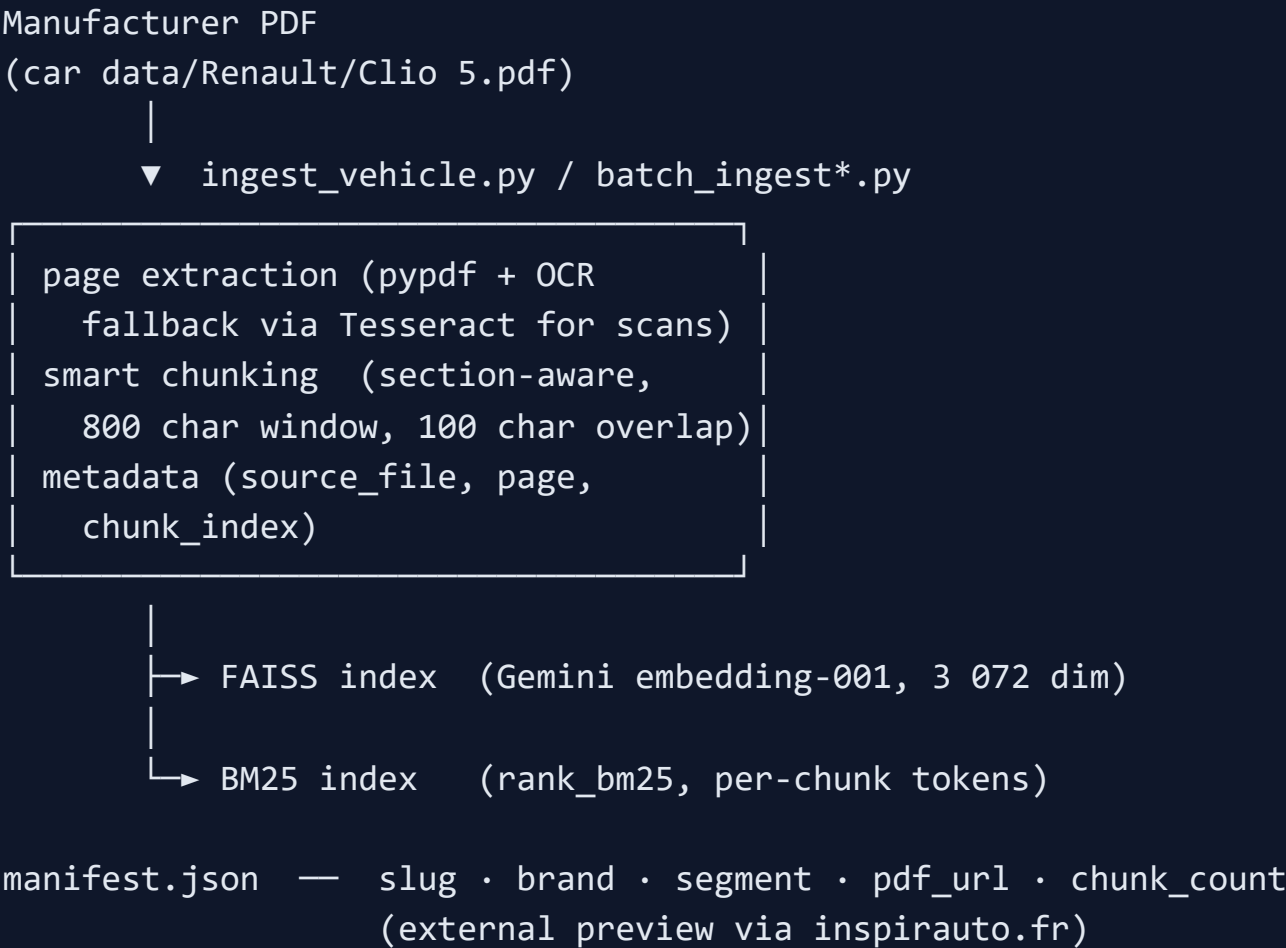
1 · System overview

Wallside turns **154 vehicle owner's manuals + 12 supplementary sub-system manuals** (infotainment, multimedia, quick-reference, maintenance logs) into a multilingual conversational assistant. A user picks a vehicle and asks a technical question (maintenance, warning lights, specifications, procedures); the system replies in French, English or Korean, grounded on the manual text and cross-checked against official manufacturer sites.

Two-stage tool-calling agent



2 · Data pipeline — how the 166 knowledge sources were indexed



Numbers

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Metric	Value
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3 · Agent tools — what the planner can call

search_manual — hybrid retrieval, per-guide

Wrapper over `_hybrid_search`. Returns up to `TOP_K_RESULTS` chunks with full metadata; the responder sees them as an EVIDENCE block with clear `[manual source.pdf p.N]` headers.

search_web — multi-site, multi-lingual, official-domain aware

The planner writes a generic query; the dispatcher **fires two parallel DDGS queries** — the raw query **and** a

`site:<manufacturer-domain>` variant built from a curated 36-brand map (audi.fr, bmw.com, hyundai.co.kr, renault.com...).

```
_OFFICIAL_DOMAINS = {  
    "audi":    ["audi.com", "audi.fr", "audiusa.com", "audi.co.kr"],  
    "bmw":     ["bmw.com", "bmw.fr", "bmwusa.com", "bmw.co.uk", ...],  
    "hyundai": ["hyundai.com", "hyundai.fr", "hyundai.co.kr", ...],  
    # 33 more  
}
```

4 · Safety · Grounding · Production

Three-layer safety

1. **Input triage** (pre-LLM, 0 ms). Accent-folded regex catches credential extraction (`GOOGLE_API_KEY` , `secret` , `.env`), prompt injection ("ignore previous instructions", role-play jailbreaks), infra probing, SSRF / path traversal, and safety-system tampering requests (disabling airbags, ECU rollback). Hard-blocked with a localised refusal — zero tokens spent.
2. **Tool-output sanitisation**. HTML tags, `javascript:` URIs, and embedded "ignore previous instructions" phrases stripped from every web / manual snippet before it reaches the responder.
3. **Model guardrail**. A `SAFETY_GUARDRAIL` system string appended to planner + responder prompts forbids revealing internal config and instructs the model to treat evidence strictly as data.

5 · Results & research questions

Live example — *"Quelle est la capacité de remorquage de l'Audi Q5 2019 ?"*

Before (linear RAG): *"According to CarsCounsel.com, 4 400 lbs. Information for 2018–2020 is not available."* — single aggregator, imperial only, fake gap.

After (agentic): *"La capacité de remorquage freinée de l'Audi Q5 (2017–2020) est de **2 000 kg (4 400 lbs)** pour une pente allant jusqu'à 12 %."* — grounded on **manual page 370**, cross-checked against **9 distinct web domains** including `media.audi.com` and `audiusa.com`, metric first.

Benchmarks

Metric	Value
Planner	700–1 200 ms
Parallel tools	250–700 ms
Responder stream (first token)	2–3 s

Thank you

Deployed: wallside.online

Source: github.com/aminssutt/AurisTraining

Contact: lakhdarberache@gmail.com

Happy to walk you through the live system, the codebase,
or any design decision in depth.